

Mahidol University
Faculty of Information
and Communication Technology

Processes

ITCS 254 | Lecture 6

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Agenda



Introduction to Processes and Process Attributes



Process Metrics and Process Control



Listing Processes



Memory Usage Monitoring



Starting Processes in the Future

...Recall...

- What is a process?
- Process types: interactive process, batch process, daemons, threads, kernel threads
- Scheduler
- Process states: running state, sleep state, zombie state

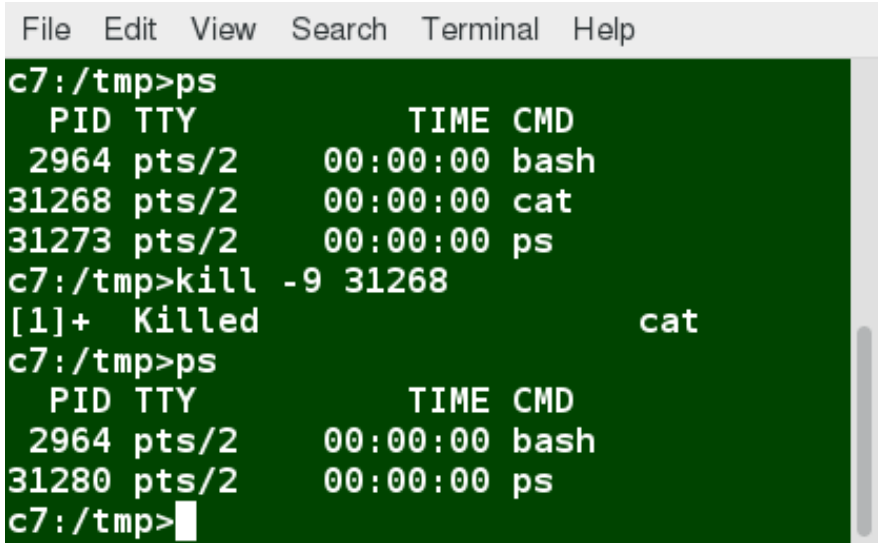
Process and Thread IDs

- The process ID (PID) is used to track process state, CPU usage, memory use, precisely where resources are located in memory, and other characteristics.
- New PIDs are usually assigned in ascending order as processes are born.
    - PID 1 denotes the `init` process (system initialization process), and succeeding processes are gradually assigned higher numbers.

ID Type	Description
Process ID (PID)	Unique Process ID number.
Parent process ID (PPID)	Process (Parent) that started this process. If the parent dies, the PPID will refer to an adoptive parent; on modern kernels, this is `kthreadd` which has PPID = 2.
Thread ID (TID)	Thread ID number. This is the same as the PID for single-threaded processes. For a multi-threaded process, each thread shares the same PID but has a unique TID.

Terminating a Process

- `kill -SIGKILL <pid>` or `kill -9 <pid>`
- You can only kill your own processes; those belonging to another user are off-limits unless you are root

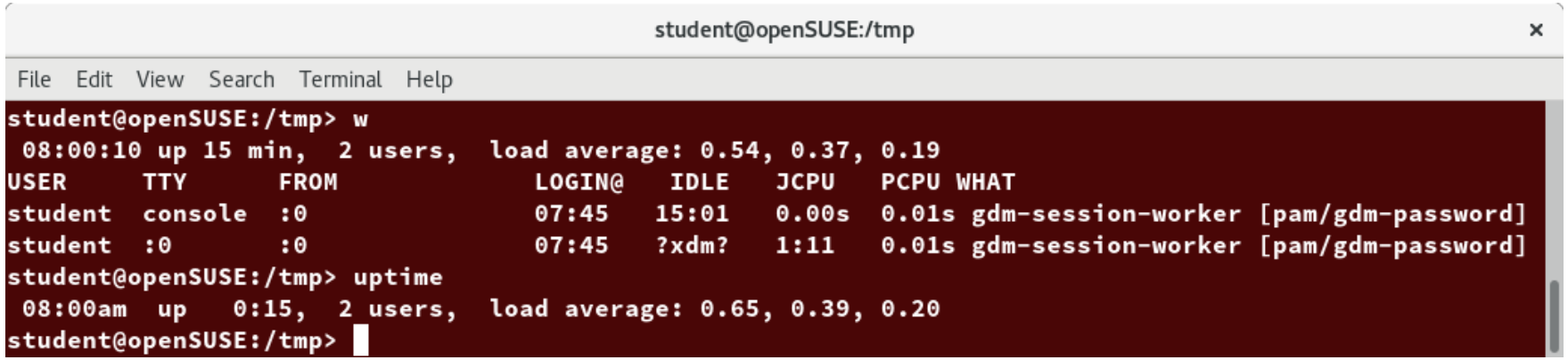


Process Priorities

- The priority for a process can be set by specifying a nice value, or niceness, for the process.
- The lower the nice value, the higher the priority.
  - In Linux, a nice value of -20 represents the highest priority and +19 represents the lowest.
  - While this may sound backwards, this convention (the nicer the process, the lower the priority) goes back to the earliest days of UNIX.
  - `renice +n <pid>` or `renice -n <pid>`
  - The `nice` command can be issued to start a process with a priority level you specify, while `renice` modifies the priority of a process that is already running.
  - `nice -n <niceness> <command>`

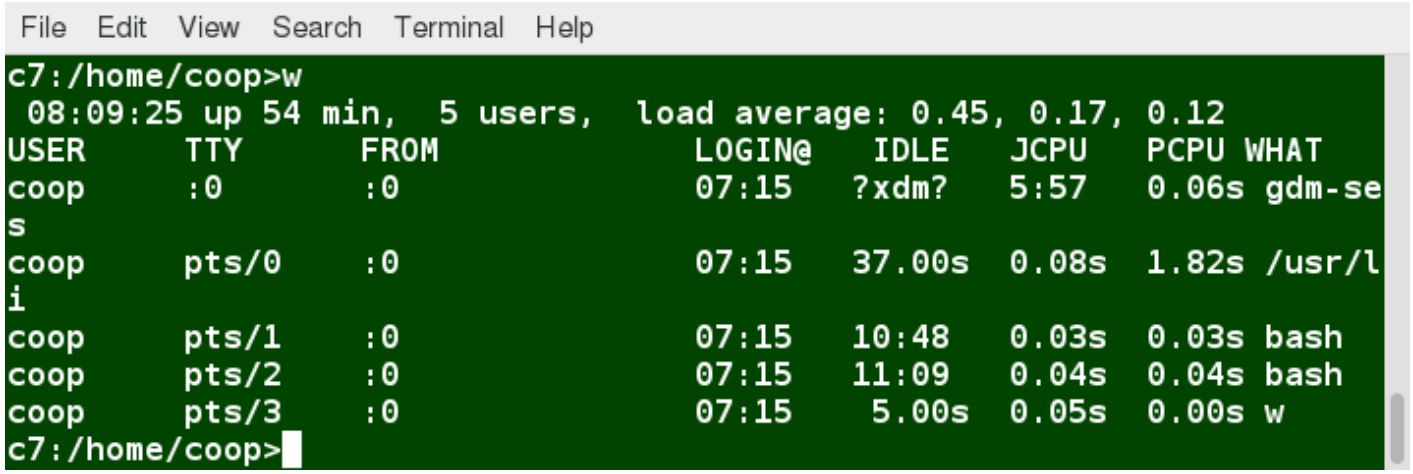
Load Averages

- The load average is the average of the load number for a given period of time.
- It takes into account processes that are:
  - Actively running on a CPU.
  - Considered runnable, but waiting on the run queue for a CPU to become available.
  - Sleeping: i.e. waiting for some kind of resource (typically, I/O) to become available.
- Linux differs from other UNIX-like OSs in that it includes the sleeping processes. Furthermore, it only includes so-called uninterruptible sleepers, those which cannot be awakened easily.



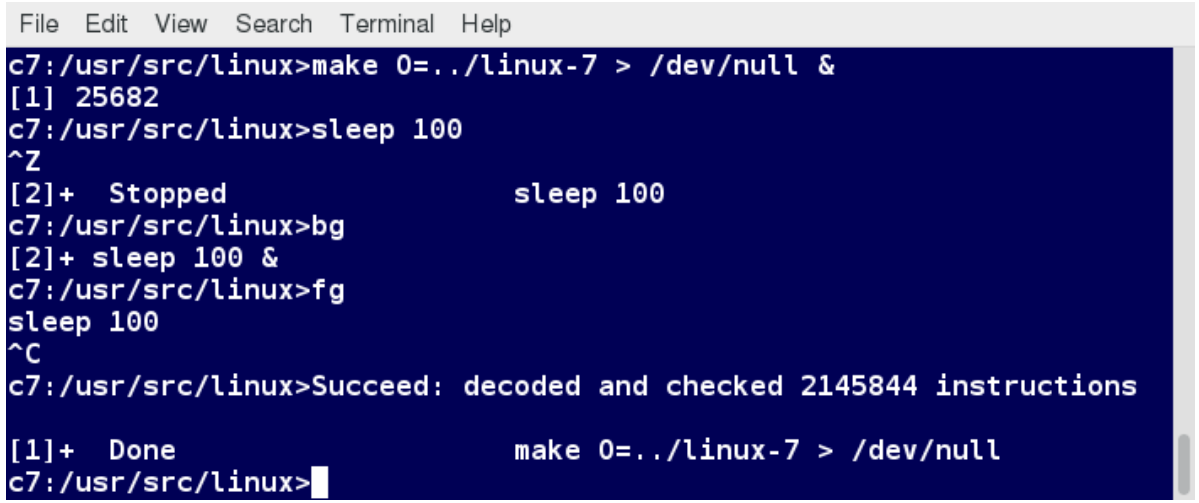
Interpreting Load Averages

- If we saw a value of 1.00 in the second position, that would imply that the single-CPU system was 100% utilized, on average, over the past 5 minutes.
- If we had more than one CPU, say a quad-CPU system, we would divide the load average numbers by the number of CPUs.
  - In this case, for example, seeing a 1 minute load average of 4.00 implies that the system as a whole was 100% (4.00/4) utilized during the last minute.



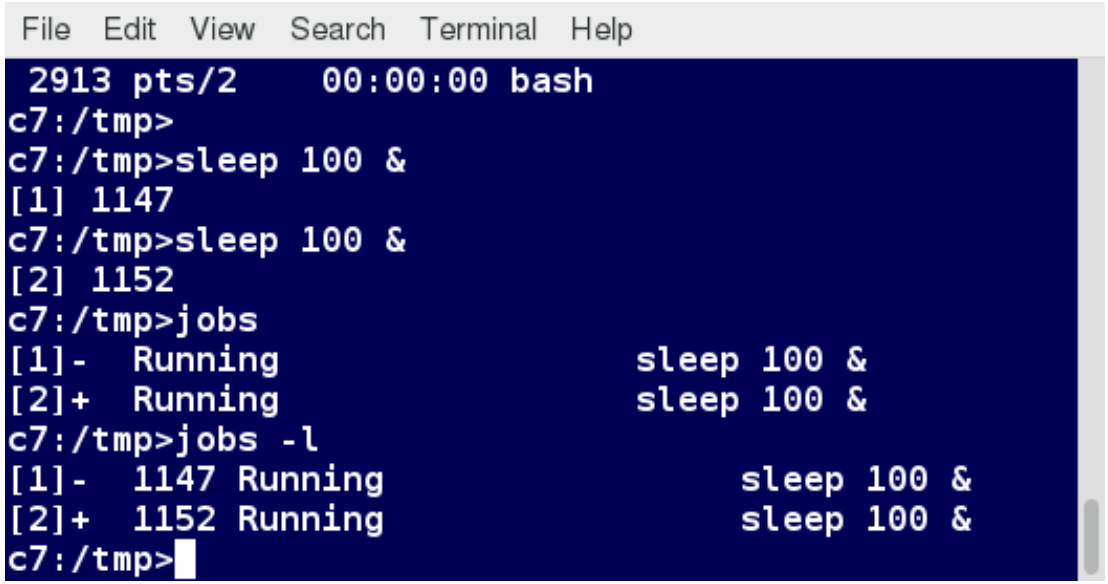
Background and Foreground Processes

- Foreground jobs run directly from the shell, and when one foreground job is running, other jobs need to wait for shell access (at least in that terminal window if using the GUI) until it is completed.
- In such cases, you can run the job in the background and free the shell for other tasks.
- The background job will be executed at lower priority, which, in turn, will allow smooth execution of the interactive tasks, and you can type other commands in the terminal window while the background job is running.
- You can use `CTRL-Z` to suspend a foreground job (i.e., put it in background) and `CTRL-C` to terminate it.



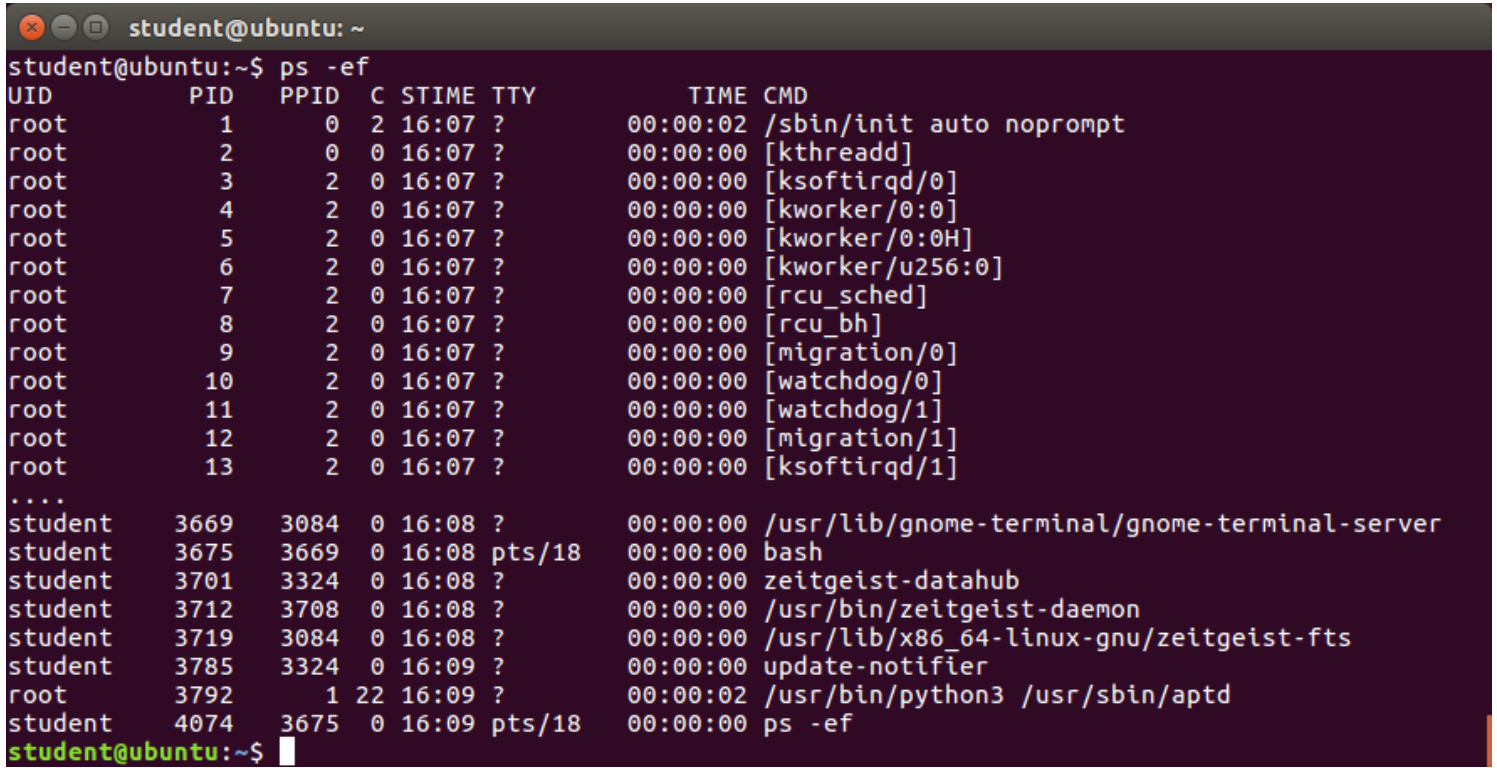
Managing Jobs

- The `jobs` utility displays all jobs running in background.
- The display shows the job ID, state, and command name.
- The background jobs are connected to the terminal window, so, if you log off, the jobs utility will not show the ones started from that window.



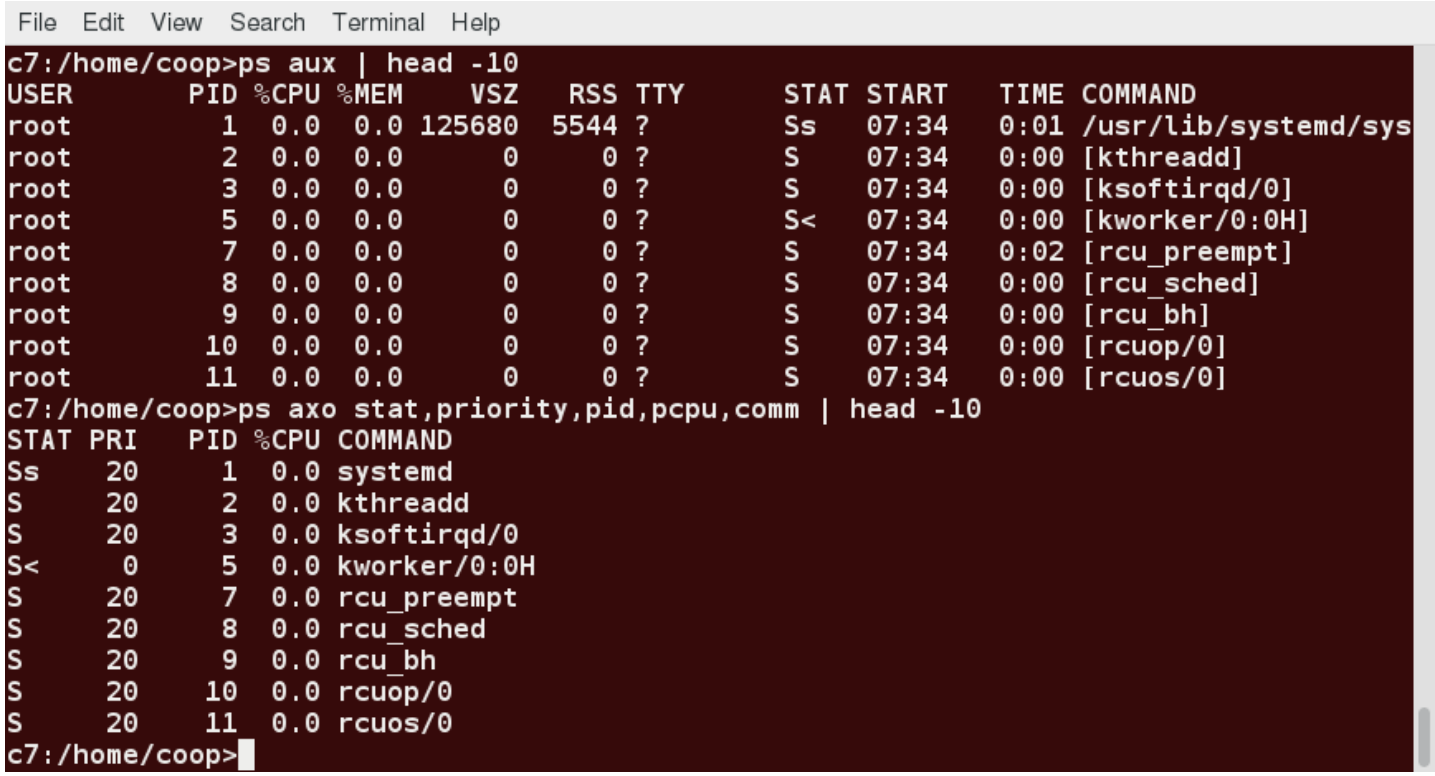
The ps Command (System V Style)

- `ps -ef` displays all the processes in the system in full detail.
- `ps -eLf` goes one step further and displays one line of information for every thread.
- `ps -l` shows priority

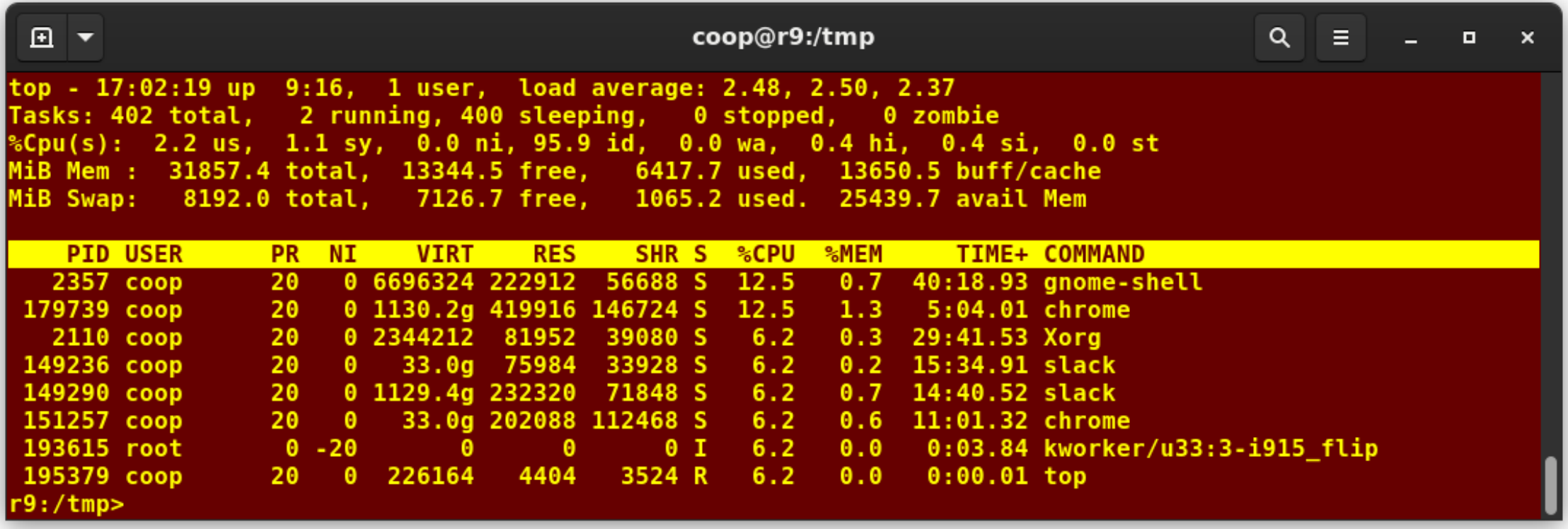


The ps Command (BSD Style)

- `ps aux` displays all processes of all users.
- `ps axo` allows you to specify which attributes you want to view.



top


coop@r9:/tmp
top - 17:02:19 up 9:16, 1 user, load average: 2.48, 2.50, 2.37
Tasks: 402 total, 2 running, 400 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.2 us, 1.1 sy, 0.0 ni, 95.9 id, 0.0 wa, 0.4 hi, 0.4 si, 0.0 st
MiB Mem : 31857.4 total, 13344.5 free, 6417.7 used, 13650.5 buff/cache
MiB Swap: 8192.0 total, 7126.7 free, 1065.2 used. 25439.7 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
2357 coop 20 0 6696324 222912 56688 S 12.5 0.7 40:18.93 gnome-shell
179739 coop 20 0 1130.2g 419916 146724 S 12.5 1.3 5:04.01 chrome
2110 coop 20 0 2344212 81952 39080 S 6.2 0.3 29:41.53 Xorg
149236 coop 20 0 33.0g 75984 33928 S 6.2 0.2 15:34.91 slack
149290 coop 20 0 1129.4g 232320 71848 S 6.2 0.7 14:40.52 slack
151257 coop 20 0 33.0g 202088 112468 S 6.2 0.6 11:01.32 chrome
193615 root 0 -20 0 0 0 I 6.2 0.0 0:03.84 kworker/u33:3-i915_flip
195379 coop 20 0 226164 4404 3524 R 6.2 0.0 0:00.01 top
r9:/tmp>

Interpreting `top` Command: Header

- Line 3
  - CPU time is being divided between the users (`us`) and the kernel (`sy`)
  - The percentage of user jobs running at a lower priority (niceness - `ni`) is then listed.
  - Idle mode (`id`) should be low if the load average is high, and vice versa.
  - The percentage of jobs waiting (`wa`) for I/O is listed.
  - Interrupts include the percentage of hardware (`hi`) vs. software interrupts (`si`).
  - Steal time (`st`) is generally used with virtual machines, which has some of its idle CPU time taken for other uses.
- Line 4 displays physical memory (RAM).
- Line 5 displays swap space.

Interpreting `top` Command: Process List

- Process Identification Number (`PID`)
- Process owner (`USER`)
- Priority (`PR`) and nice values (`NI`)
- Virtual (`VIRT`), physical (`RES`), and shared memory (`SHR`)
- Status (`S`): `S` = interruptible sleep, `D` = uninterruptible sleep, `I` = idle, and `R` = running
- Percentage of CPU (`%CPU`) and memory (`%MEM`) used
- Execution time (`TIME+`)
- Command (`COMMAND`)

Interactive Keys with `top`

Command	Output
`h` or `?`	Display available interactive keys and their function.
`t`	Display or hide summary information (rows 2 and 3).
`m`	Display or hide memory information (rows 4 and 5).
`1`	Show information for each CPU and not just totals.
`d`	Change display update interval.
`A`	Sort the process list by top resource consumers.
`r`	Renice (change the priority of) a specific processes.
`k`	Kill a specific process.
`f`	Enter the `top` configuration screen.
`o`	Interactively select a new sort order in the process list.

atop

```
ATOP - linuxfordevices                          2020/06/26  20:58:28                          --------------                          10s elapsed
PRC | sys    1.61s | user   7.71s |  #proc    315 | #trun      1 | #tslpi   820 | #tslpu     0 | #zombie    0 | clones    82 | #exit     24 |
CPU | sys      19% | user     79% |  irq       3% | idle    292% | wait      8% | steal     0% | guest     0% | curf  798MHz | curscal  42% |
cpu | sys       5% | user     22% |  irq       0% | idle     72% | cpu002 w  0% | steal     0% | guest     0% | curf  798MHz | curscal  42% |
cpu | sys       3% | user     21% |  irq       0% | idle     70% | cpu000 w  5% | steal     0% | guest     0% | curf  798MHz | curscal  42% |
cpu | sys       6% | user     17% |  irq       1% | idle     74% | cpu001 w  1% | steal     0% | guest     0% | curf  798MHz | curscal  42% |
cpu | sys       4% | user     18% |  irq       1% | idle     75% | cpu003 w  1% | steal     0% | guest     0% | curf  798MHz | curscal  42% |
CPL | avg1    1.44 | avg5    0.83 |  avg15   0.89 |               | csw    45740 |               | intr   20758 |              | numcpu     4 |
MEM | tot     7.7G | free    3.4G |  cache   1.9G | buff  152.4M | slab  152.0M | shmem 678.3M | vmbal   0.0M | hptot   0.0M | hpuse   0.0M |
SWP | tot     2.0G | free    2.0G |               |               |               |               |              | vmcom   9.4G | vmlim   5.8G |
DSK |          sda | busy      4% |  read    183 | write    145 | KiB/r     10 | KiB/w     13 | MBr/s    0.2 | MBw/s    0.2 | avio 1.13 ms |
NET | transport     | tcpi      0 |  tcpo      0 | udpi      23 | udpo      11 | tcpao      0 | tcppo      0 | tcprs      7 | udpie      0 |
NET | network       | ipi      11 |  ipo      18 | ipfrw      0 | deliv     11 |               |              | icmpi      0 | icmpo      0 |
NET | wlp19s0   0%  | pcki      2 |  pcko     15 | sp   72 Mbps | si    0 Kbps | so    1 Kbps | erri       0 | erro       0 | drpo       0 |
NET | lo      ----  | pcki      3 |  pcko      3 | sp    0 Mbps | si    0 Kbps | so    0 Kbps | erri       0 | erro       0 | drpo       0 |

  PID   TID S  CPU COMMAND-LINE (horizontal scroll with <- and -> keys)                                                                1/5
 1877     - S  27% /usr/bin/gnome-shell
 2282     - S  12% /opt/google/chrome/chrome --type=gpu-process --field-trial-handle=3127800387498089770,9196475103215302438,131072 --gpu-pref
 1731     - S  12% /usr/lib/xorg/Xorg vt2 -displayfd 3 -auth /run/user/1000/gdm/Xauthority -background none -noreset -keeptty -verbose 3
 2568     - S  11% /opt/google/chrome/chrome --type=renderer --disable-webrtc-apm-in-audio-service --field-trial-handle=3127800387498089770,91
 7196     - E   5% locate
 7161     - S   5% /usr/bin/nautilus --gapplication-service
 7165     - S   3% /usr/bin/seahorse --no-window
 7164     - S   3% /usr/bin/gnome-calendar --gapplication-service
 6102     - S   3% /opt/google/chrome/chrome --type=renderer --disable-webrtc-apm-in-audio-service --field-trial-handle=3127800387498089770,91
 7226     - S   3% /usr/bin/gnome-screenshot --gapplication-service
 7159     - S   1% /usr/lib/gnome-control-center/gnome-control-center-search-provider
 2211     - S   1% /opt/google/chrome/chrome
 2235     - S   1% /usr/lib/gnome-terminal/gnome-terminal-server
 1749     - S   1% /usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
 7156     - R   1% atop -c
 2315     - S   0% /opt/google/chrome/chrome --type=renderer --disable-webrtc-apm-in-audio-service --field-trial-handle=3127800387498089770,91
 2286     - S   0% /opt/google/chrome/chrome --type=utility --field-trial-handle=3127800387498089770,9196475103215302438,131072 --lang=en-GB -
    1     - S   0% /sbin/init splash
 1916     - S   0% ibus-daemon --xim --panel disable
 4887     - S   0% /opt/google/chrome/chrome --type=renderer --disable-webrtc-apm-in-audio-service --field-trial-handle=3127800387498089770,91
 2155     - S   0% /usr/lib/evolution/evolution-calendar-factory-subprocess --factory all --bus-name org.gnome.evolution.dataserver.Subproces
```

Image: https://www.digitalocean.com/community/tutorials/atop-command-in-linux

btop

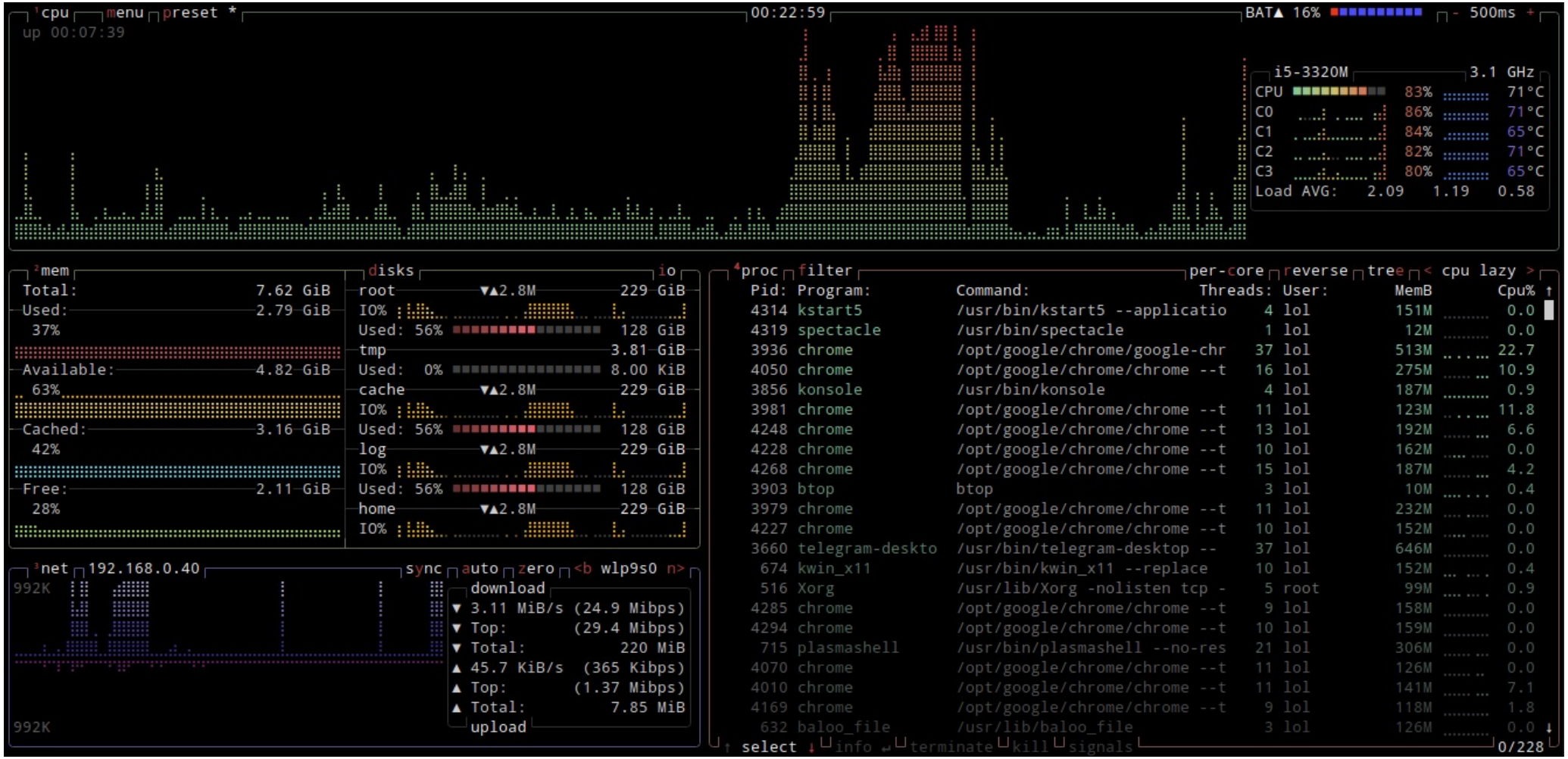
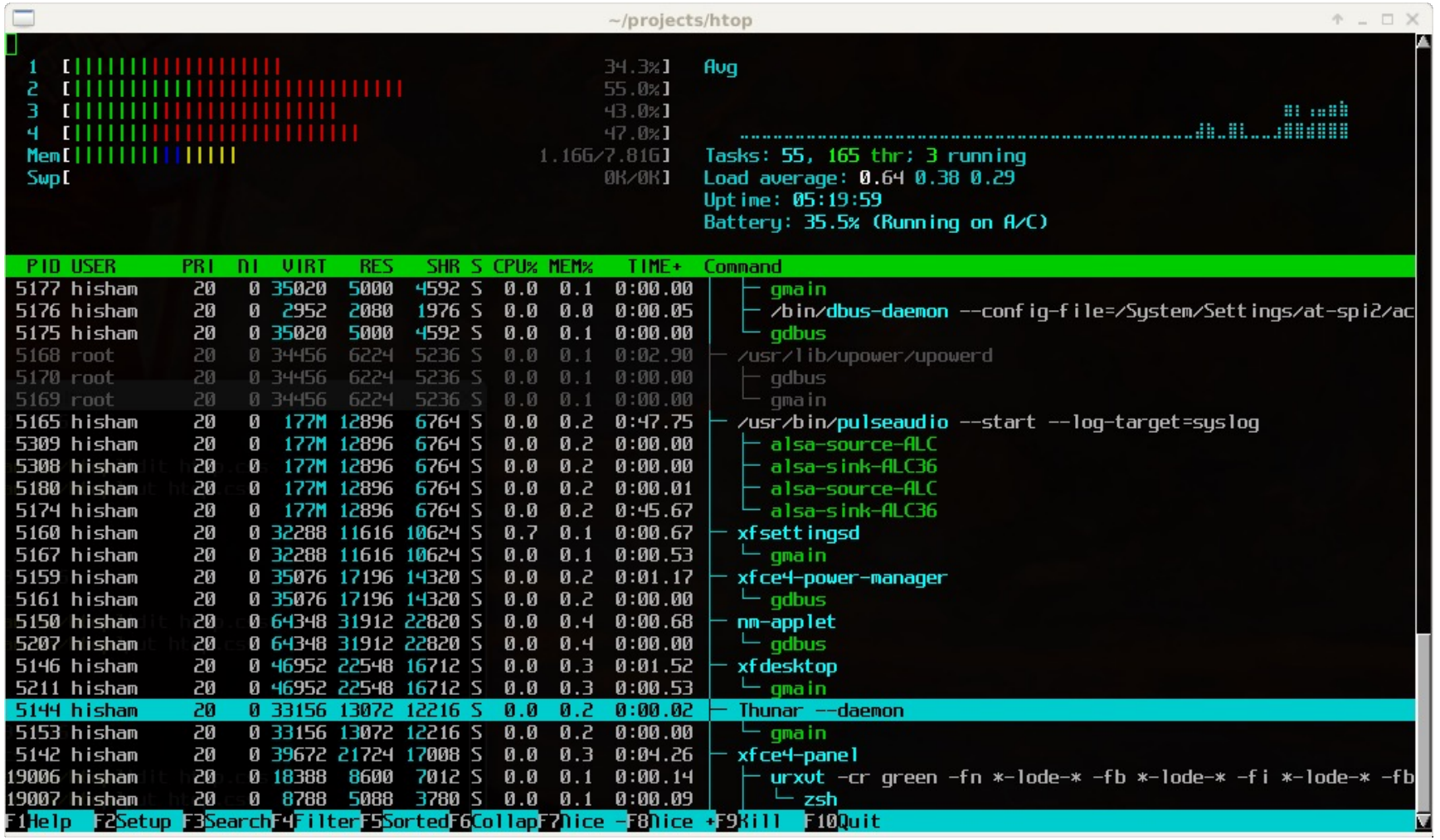


Image: https://en.m.wikipedia.org/wiki/File:Btop_screenshot.png

htop



https://htop.dev

Memory Usage

```
$ top -o %MEM
```

```
top - 02:10:32 up 57 min,  0 users,  load average: 0.00, 0.01, 0.00
Tasks: 120 total,   1 running, 119 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.3 us,  0.3 sy,  0.0 ni, 99.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1983.3 total,    245.6 free,    314.4 used,   1423.3 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   1476.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2669 root      20   0  522200  89724  27528 S   0.0   4.4   0:00.71 fwupd
    759 root      20   0 1764860  81144  55080 S   0.0   4.0   0:00.37 dockerd
   1145 root      39  19  848532  68268  33288 S   0.0   3.4   0:01.56 node
   2798 root      39  19  832004  50948  29496 S   0.0   2.5   0:01.69 node
    573 root      20   0 1727048  45020  32412 S   0.0   2.2   0:00.43 containerd
   1556 root      20   0 1321432  32120  20596 S   0.0   1.6   0:04.00 snapd
    747 root      20   0  108120  20916  13252 S   0.0   1.0   0:00.09 unattended-u+
    554 root      20   0   29864  18604  10332 S   0.0   0.9   0:00.07 networkd-dis+
    474 root      rt   0  280200  18000   8208 S   0.0   0.9   0:00.14 multipathd
      1 root      20   0  103876  12980   8520 S   0.0   0.6   0:02.35 systemd
    352 root      19  -1   35748  12692  11596 S   0.0   0.6   0:00.16 systemd-jour+
    895 systemd+  20   0   24440  12332   8432 S   0.0   0.6   0:00.03 systemd-reso+
   1161 root       0 -20 1231180  12152   5468 S   0.0   0.6   0:03.35 runtime-info+
    565 root      20   0  393060  12148  10316 S   0.0   0.6   0:00.04 udisksd
    653 root      20   0  315100  11396   9684 S   0.0   0.6   0:00.05 ModemManager
    557 root      20   0  234120   8476   6576 S   0.0   0.4   0:00.05 polkitd
    385 systemd+  20   0   27264   7564   6696 S   0.0   0.4   0:00.03 systemd-netw+
    562 root      20   0   17232   7464   6588 S   0.0   0.4   0:00.17 systemd-logi+
    542 root      20   0  237324   7424   6596 S   0.0   0.4   0:00.05 accounts-dae+
    652 root      20   0   12180   7316   6420 S   0.0   0.4   0:00.00 sshd
```

Memory Usage

```
ubuntu $ free
               total        used        free      shared  buff/cache   available
Mem:         2030940      311656      265600        1020     1453684     1522136
Swap:        1048572           0     1048572
ubuntu $ free -mt
               total        used        free      shared  buff/cache   available
Mem:            1983         304         259           0        1419        1486
Swap:           1023           0        1023
Total:          3007         304        1283
ubuntu $
```

```
ubuntu $ cat /proc/meminfo
MemTotal:        2030940 kB
MemFree:          250732 kB
MemAvailable:    1511112 kB
Buffers:          113272 kB
Cached:          1179844 kB
SwapCached:            0 kB
Active:           946696 kB
Inactive:         566596 kB
Active(anon):     229188 kB
Inactive(anon):      208 kB
Active(file):     717508 kB
Inactive(file):   566388 kB
Unevictable:       18536 kB
Mlocked:           18536 kB
SwapTotal:       1048572 kB
SwapFree:        1048572 kB
Dirty:                 0 kB
Writeback:             0 kB
AnonPages:        238720 kB
Mapped:           208516 kB
Shmem:              1020 kB
KReclaimable:     164412 kB
Slab:             215064 kB
SReclaimable:     164412 kB
SUnreclaim:        50652 kB
KernelStack:        3308 kB
PageTables:         4092 kB
```

Virtual Memory Statistics: `vmstat`

```
ubuntu $ vmstat -help

Usage:
 vmstat [options] [delay [count]]

Options:
 -a, --active           active/inactive memory
 -f, --forks            number of forks since boot
 -m, --slabs            slabinfo
 -n, --one-header       do not redisplay header
 -s, --stats            event counter statistics
 -d, --disk             disk statistics
 -D, --disk-sum         summarize disk statistics
 -p, --partition <dev>  partition specific statistics
 -S, --unit <char>      define display unit
 -w, --wide             wide output
 -t, --timestamp        show timestamp

 -h, --help     display this help and exit
 -V, --version  output version information and exit

For more details see vmstat(8).
```

```
ubuntu $ vmstat -s
      2030940 K total memory
       318112 K used memory
       941396 K active memory
       567084 K inactive memory
       255592 K free memory
       113204 K buffer memory
      1344032 K swap cache
      1048572 K total swap
            0 K used swap
      1048572 K free swap
         1554 non-nice user cpu ticks
          542 nice user cpu ticks
         1133 system cpu ticks
       311914 idle cpu ticks
         1931 IO-wait cpu ticks
            0 IRQ cpu ticks
           60 softirq cpu ticks
          590 stolen cpu ticks
       839016 pages paged in
       831298 pages paged out
            0 pages swapped in
            0 pages swapped out
       373402 interrupts
       890041 CPU context switches
   1738804406 boot time
         2936 forks
ubuntu $
```

```
ubuntu $ vmstat -D
            9 disks
            3 partitions
        49574 total reads
         5409 merged reads
      1678032 read sectors
        19235 milli reading
         6322 writes
        24662 merged writes
      1662709 written sectors
        75072 milli writing
            0 inprogress IO
           35 milli spent IO
ubuntu $
```

System Monitoring Using `vmstat`

```
ubuntu $ vmstat -a 2 1000
procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
 r  b   swpd   free  inact active   si   so    bi    bo   in   cs us sy id wa st
 0  0      0 330384 565556 869764    0    0  7851  7682 1591 4000 18 10 54 15  3
 0  0      0 329408 565544 870164    0    0     0     0  134  371  5  1 95  0  0
 0  0      0 330164 565544 869452    0    0     0     0  120  275  0  0 100  0  0
 0  0      0 330164 565544 869648    0    0     0     8  124  278  0  0 99  0  0
 0  0      0 330132 565544 869876    0    0     0     0  115  266  1  0 99  0  0
 0  0      0 329880 565544 870068    0    0     0  3332  215  295  0  0 95  4  0
 0  0      0 329872 565544 870232    0    0     0     0  107  240  0  0 100  0  0
 5  0      0 329644 565544 869740    0    0     2   618  354  843  2  1 82 17  0
 0  0      0 335164 562220 868832    0    0     0   254  216 1788  7  5 85  3  0
 0  0      0 335416 562200 868252    0    0     0     0  133  350  5  1 94  0  0
 0  0      0 335416 562200 868468    0    0     0    12  118  255  0  0 99  1  0
 0  0      0 335384 562200 868652    0    0     0     0  108  224  0  0 100  0  0
 0  0      0 335384 562200 868888    0    0     0     0  115  242  1  0 99  0  0
 0  0      0 335384 562200 869104    0    0     0     0  109  238  0  0 100  0  0
 0  0      0 335132 562200 868616    0    0     0     0  104  222  0  0 99  0  0
 0  0      0 335132 562200 868804    0    0     0     0  116  246  0  0 100  0  0
 0  0      0 335164 562200 869076    0    0     0     0  105  222  1  0 100  0  0
 0  0      0 335668 562200 868148    0    0     0     0  115  245  0  1 100  0  0
 0  0      0 335668 562200 868352    0    0     0     0  110  238  0  0 100  0  0
 0  0      0 335668 562200 868512    0    0     0     0  102  219  0  0 100  0  1
 0  0      0 335888 562200 867940    0    0     0     0  114  246  0  0 100  0  0
 0  0      0 335888 562200 868148    0    0     0     0  109  231  0  0 99  0  0
```

- `r` = runnable proc (waiting for CPU time)
- `b` = blocked proc (waiting for I/O operations)
- `si` = swap-in-rate (KB/s data moved from swap to RAM)
- `so` = swap-out-rate (moved from RAM to swap
- `bi` = disk reads
- `bo` = disk writes
- `in` = interrupts per second
- `cs` = context switches per second
- `us` = user time
- `sy` = system time
- `id` = idle time
- `wa` = i/o wait time
- `st` = steal time in a virtualized environment

Scheduling Future Process Using `at`

```
c8:/tmp>at now + 2 days
warning: commands will be executed using /bin/sh

at> # now issue a command
at> mail < /var/log/messages student@localhost

at> # hit CTRL-D to quit
at> <EOT>
job 103 at Sun Jun  6 15:43:00 2021

c8:/tmp># check on scheduled job:
c8:/tmp>atq
103     Sun Jun  6 15:43:00 2021 a coop

c8:/tmp># remove the scheduled job
c8:/tmp>atrm 103

c8:/tmp>#check
c8:/tmp>atq
c8:/tmp>
```

Alternatively, you can run a script file: `at now + 1 minute -f script.sh`

cron

- `cron` is a time-based scheduling utility program.
- It can launch routine background jobs at specific times and/or days on an ongoing basis.
- `cron` is driven by a configuration file called `/etc/crontab` (cron table)
- There are both system-wide crontab files and individual user-based ones.
- Each line of a crontab file represents a job, and is composed of a so-called CRON expression, followed by a shell command to execute.
- `crontab -e` will open the crontab editor to edit existing jobs or to create new jobs.
  - Alternatively, you can create a file containing scheduled jobs, then run command `crontab <filename>`
- `crontab -l` is to list all scheduled jobs.
- `crontab -r` is to remove jobs.

`crontab` File

Field	Description	Values
`MIN`	Minutes	0 - 59
`HOUR`	Hour field	0 - 23
`DOM`	Day of month	1 - 31
`MON`	Month field	1 - 12
`DOW`	Day of week	0 - 6 (0 = Sunday)
`CMD`	Command	Any command to be executed

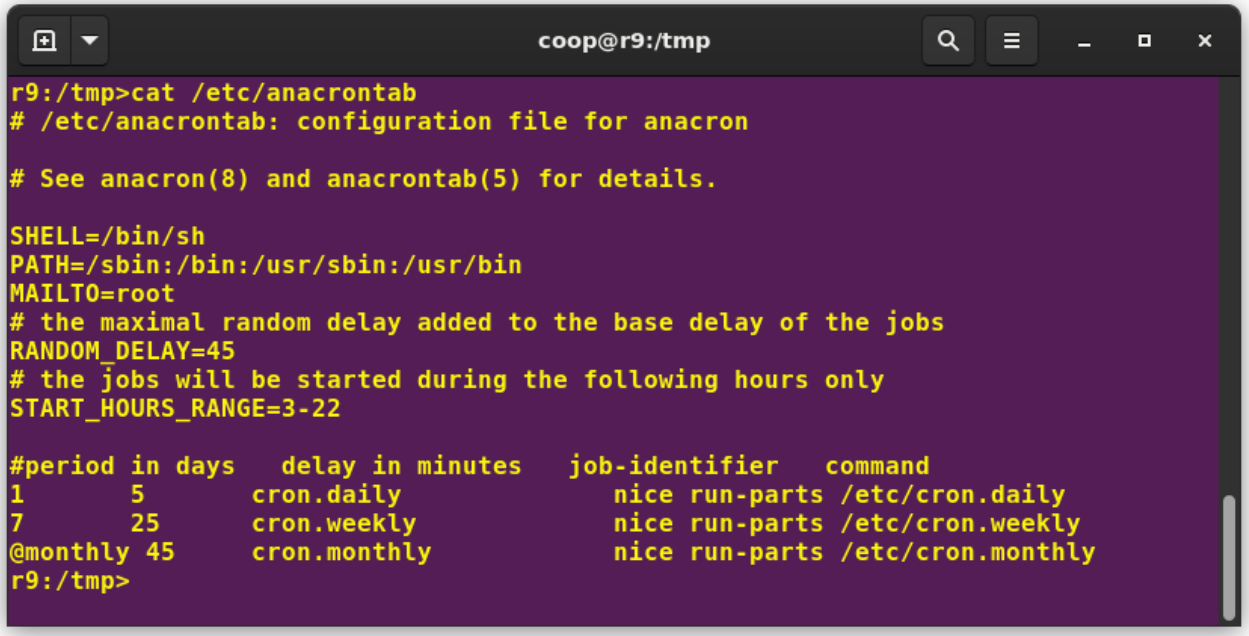
Examples:

```
* * * * * /usr/local/bin/execute/this/script.sh

30 08 10 06 * /home/sysadmin/full-backup
```

anacron

- `anacron` is a newer facility.
- This was because `cron` implicitly assumed the machine was always running.
- If the machine was powered off, scheduled jobs would not run. `anacron` will run the necessary jobs in a controlled and staggered manner when the system is up and running.



sleep

- Sometimes, a command or job must be delayed or suspended.
- `sleep` suspends execution for at least the specified period of time, which can be given as the number of seconds (`s`, the default), minutes (`m`), hours (`h`), or days (`d`).
- After that time has passed (or an interrupting signal has been received), execution will resume.

